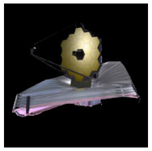

7199 - Anomalous infrared fluxes of M dwarfs

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Erik Zackrisson (PI) (ESA Member)	**Uppsala Astronomical Observatory**
Dr. Uma Gorti (CoI)	SETI Institute
Dr. Jason T Wright (CoI) (US Admin CoI)	The Pennsylvania State University
Dr. Roberto Assef (CoI)	Universidad Diego Portales
Andrew Blain (CoI) (ESA Member)	University of Leicester
Michael A. Garrett (CoI) (ESA Member)	University of Manchester Institute of Science and Technology
Tongtian Ren (CoI) (ESA Member)	University of Manchester
Dr. Arjan Bik (CoI) (ESA Member)	Stockholm University
Dr. Armin Nabizadeh (CoI) (ESA Member)	Uppsala University
Dr. Alexis Brandeker (CoI) (ESA Member)	Stockholm University
Prof. Andrew Siemion (CoI)	University of California - Berkeley

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	4	Object_A_background	MIRI Medium Resolution Spectroscopy	(2) Object_A_background
	5	Object_D_background	MIRI Medium Resolution Spectroscopy	(7) Object_D_background
	6	Object_E_background	MIRI Medium Resolution Spectroscopy	(8) Object_E_background
	1	Object_A	MIRI Medium Resolution Spectroscopy	(1) Object_A
	2	Object_D	MIRI Medium Resolution Spectroscopy	(3) Object_D
	3	Object_E	MIRI Medium Resolution Spectroscopy	(4) Object_E

ABSTRACT

We target three very-unusual "one-in-a-million" main-sequence M dwarfs that show extreme levels of mid-infrared excess flux. These objects were singled out from a sample of five million stars sampled by Gaia and WISE in a search motivated by the quest to detect technosignatures, or Dyson spheres. Regardless of the physical explanation for the mid-infrared excess, JWST's unique combination of sensitivity, spectral range and spectral and spatial resolution will allow different explanations for the excess to be tested.

Three possible explanations have been proposed for the remarkable mid-IR excess of these M dwarfs:
1) An extreme, late-forming debris disk due to collisions within their planetary systems;
2) Contamination by a rare very-luminous type of WISE-discovered obscured active galactic nucleus that only contributes significantly to the flux at wavelengths longer than 10 micron, and which happens to be projected within a few arcseconds of the M dwarf;
3) A bona-fide Dyson sphere absorbing starlight, and operating at ~100-200 K.

We propose a combination of MRS spectroscopy at 5-25 micron, along with imaging in F560W, F1000W and F1500W of the surrounding field to uncover the true nature of these objects.

This project has the potential to make: the first detection of extreme debris disks without signs of ongoing gas accretion around mature M dwarfs; a "black swan" detection of an alien civilization; or to reveal three fascinating distant galaxies masked by a foreground star, which will provide precious adaptive-optics-friendly targets for future ELT spectroscopy.

OBSERVING DESCRIPTION

This programme will carry out MRS spectroscopy at 5-25 micron (channels 1-4, wavelength ranges A, B and C) of three M dwarfs with extreme levels of infrared excess flux. While dedicated MRS background observations are obtained in direct connection to the target observations, we will also carry out MIRI imaging in filters F560W, F1000W and F1500W of the field around the targets to characterize the surroundings and search for faint structures or contaminating objects close to the targets.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Object_A	RA: 12 45 12.9485 (191.3039521d) Dec: -26 52 1.86 (-26.86718d) Equinox: J2000	Proper Motion RA: -5.717 mas/yr Proper Motion Dec: -88.75999992596917 mas/yr Parallax: 0.0069341" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[M dwarfs]

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	Object_A_background	RA: 12 45 12.9485 (191.3039521d) Dec: -26 52 1.86 (-26.86718d) Equinox: J2000	Proper Motion RA: -5.717 mas/yr Proper Motion Dec: -88.75999992596917 mas/yr Parallax: 0.0069341" Epoch of Position: 2000	

Comments:
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	Object_D	RA: 23 27 51.3277 (351.9638654d) Dec: +05 06 26.49 (5.10736d) Equinox: J2000	Proper Motion RA: -30.674 mas/yr Proper Motion Dec: -21.61099996556004 mas/yr Parallax: 0.004695700000000001" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[M dwarfs]

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	Object_E	RA: 04 02 7.7601 (60.5323338d) Dec: -10 54 41.21 (-10.91145d) Equinox: J2000	Proper Motion RA: 33.957 mas/yr Proper Motion Dec: 30.112 mas/yr Parallax: 0.0036233" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[M dwarfs]

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(7)	Object_D_background	RA: 23 27 52.7400 (351.9697500d) Dec: +05 06 41.60 (5.11156d) Equinox: J2000	Proper Motion RA: -30.674 mas/yr Proper Motion Dec: -21.61099996556004 mas/yr Parallax: 0.004695700000000001" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

(8)	Object_E_background	RA: 04 02 7.7601 (60.5323338d) Dec: -10 54 41.21 (-10.91145d) Equinox: J2000	Proper Motion RA: 33.957 mas/yr Proper Motion Dec: 30.112 mas/yr Parallax: 0.0036233" Epoch of Position: 2000

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

Observation

Proposal 7199, Observation 4: Object_A_background Mon Jun 23 16:00:10 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Object_A (Obs 1)]

Comments: Special requirements on VP3A are used to avoid the MRS falling onto bright WISE sources

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	Object_A_background	RA: 12 45 12.9485 (191.3039521d) Dec: -26 52 1.86 (-26.86718d) Equinox: J2000	Proper Motion RA: -5.717 mas/yr Proper Motion Dec: -88.75999992596917 mas/yr Parallax: 0.0069341" Epoch of Position: 2000	

Comments:
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F1500W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.1
1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
2		IMAGER	F1000W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.1
2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
3		IMAGER	F1500W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.1
3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	

Special Requirements	Aperture PA Range 84.83544897 to 119.83544897 Degrees (V3 80.0 to 115.0) Aperture PA Range 294.83544897 to 304.83544897 Degrees (V3 290.0 to 300.0) Sequence Observations 4, 1 (reordered), Non-interruptible

Observation

Proposal 7199, Observation 5: Object_D_background Mon Jun 23 16:00:10 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Object_D (Obs 2)]

Comments: Special requirements on VP3A are used to avoid the MRS falling onto bright WISE sources

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(7)	Object_D_background	RA: 23 27 52.7400 (351.9697500d) Dec: +05 06 41.60 (5.11156d) Equinox: J2000	Proper Motion RA: -30.674 mas/yr Proper Motion Dec: -21.61099996556004 mas/yr Parallax: 0.004695700000000001" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F1500W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.4
1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
2		IMAGER	F1000W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.4
2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
3		IMAGER	F1500W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.4
3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	

Special Requirements	
	Aperture PA Range 64.83544897 to 89.83544897 Degrees (V3 60.0 to 85.0) Aperture PA Range 234.83544897 to 239.83544897 Degrees (V3 230.0 to 235.0) Sequence Observations 5, 2 (reordered), Non-interruptible

Observation

Proposal 7199, Observation 6: Object_E_background

Mon Jun 23 16:00:10 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Object_E (Obs 3)]

Comments: No special requirements on VP3A are listed, since there are no WISE sources that may affect the MRS within the range of roll angles allowed for this target

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	Object_E_background	RA: 04 02 7.7601 (60.5323338d) Dec: -10 54 41.21 (-10.91145d) Equinox: J2000	Proper Motion RA: 33.957 mas/yr Proper Motion Dec: 30.112 mas/yr Parallax: 0.0036233" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F1500W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F560W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.5
1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
2		IMAGER	F1000W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.5
2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
3		IMAGER	F1500W	FASTR1	6	14	1	Dither 1	4	56	1076.716	221637.5
3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	
3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	

Special Requirements	
	Sequence Observations 6, 3 (reordered), Non-interruptible

Observation

Proposal 7199, Observation 1: Object_A

Mon Jun 23 16:00:10 GMT 2025

Diagnostic Status: Error

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Object_A_background (Obs 4)]

Comments: The APT error appears to be error 3 (due to a bug) described here: https://stsci.service-now.com/jwst?id=kb_article_view&sys_kb_id=a14ea52adbfc360042685434ce9619eb

Diagnostics

(Object_A (Obs 1)) Error (Form): This target requires similar background exposures that are linked in a non-interruptible group/sequence.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Object_A	RA: 12 45 12.9485 (191.3039521d) Dec: -26 52 1.86 (-26.86718d) Equinox: J2000	Proper Motion RA: -5.717 mas/yr Proper Motion Dec: -88.75999992596917 mas/yr Parallax: 0.0069341" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[M dwarfs]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1500W	FAST	4	1	1	11.1	221637.3

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	NO	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.2
1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.2
2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.2
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.2
3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.2
3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.2

Special Requirements	
	Sequence Observations 4, 1 (reordered), Non-interruptible

Observation

Proposal 7199, Observation 2: Object_D Mon Jun 23 16:00:10 GMT 2025

Diagnostic Status: Error

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Object_D_background (Obs 5)]

Comments: The APT error appears to be error 3 (due to a bug) described here: https://stsci.service-now.com/jwst?id=kb_article_view&sys_kb_id=a14ea52adbfc360042685434ce9619e

Diagnostics

(Object_D (Obs 2)) Error (Form): This target requires similar background exposures that are linked in a non-interruptible group/sequence.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	Object_D	RA: 23 27 51.3277 (351.9638654d) Dec: +05 06 26.49 (5.10736d) Equinox: J2000	Proper Motion RA: -30.674 mas/yr Proper Motion Dec: -21.61099996556004 mas/yr Parallax: 0.004695700000000001" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[M dwarfs]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1500W	FAST	4	1	1	11.1	221637.8

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	NO	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.6
1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.6
2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.6
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.6
3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.6
3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.6

Special Requirements	
	Sequence Observations 5, 2 (reordered), Non-interruptible

Observation

Proposal 7199, Observation 3: Object_E Mon Jun 23 16:00:10 GMT 2025

Diagnostic Status: Error

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Object_E_background (Obs 6)]

Comments: The APT error appears to be error 3 (due to a bug) described here: https://stsci.service-now.com/jwst?id=kb_article_view&sys_kb_id=a14ea52adbfc360042685434ce9619e

Diagnostics

(Object_E (Obs 3)) Error (Form): This target requires similar background exposures that are linked in a non-interruptible group/sequence.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	Object_E	RA: 04 02 7.7601 (60.5323338d) Dec: -10 54 41.21 (-10.91145d) Equinox: J2000	Proper Motion RA: 33.957 mas/yr Proper Motion Dec: 30.112 mas/yr Parallax: 0.0036233" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[M dwarfs]

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F1500W	FAST	4	1	1	11.1	221637.9

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	NO	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	POINT SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SHORT(A)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.7
1	SHORT(A)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.7
2	MEDIUM(B)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.7
2	MEDIUM(B)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.7
3	LONG(C)	MRSLONG		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.7
3	LONG(C)	MRSSHORT		SLOWR1	12	1	1	Dither 1	4	4	1146.716	221637.7

Special Requirements	
	Sequence Observations 6, 3 (reordered), Non-interruptible